

## Questions

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1. (a) Show that, for  $r > 0$ ,

$$\ln \left( 1 + \frac{2}{r(r+3)} \right) = \ln \left( \frac{r+1}{r} \right) - \ln \left( \frac{r+3}{r+2} \right) \quad [1]$$

- (b) Hence, using the method of differences, show that for integer  $n \geq 1$ ,

$$\sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right) = \ln \left( \frac{3(n+1)}{n+3} \right) \quad [4]$$

### Solution

- (a) Since  $r > 0$ , all logarithms are defined. Also,

$$1 + \frac{2}{r(r+3)} = \frac{r(r+3)+2}{r(r+3)} = \frac{r^2+3r+2}{r(r+3)} = \frac{(r+1)(r+2)}{r(r+3)} = \frac{r+1}{r} \cdot \frac{r+2}{r+3}$$

So

$$\begin{aligned} \ln \left( 1 + \frac{2}{r(r+3)} \right) &= \ln \left( \frac{r+1}{r} \right) + \ln \left( \frac{r+2}{r+3} \right) \\ &= \ln \left( \frac{r+1}{r} \right) - \ln \left( \frac{r+3}{r+2} \right) \end{aligned}$$

which is the required result.

- (b) Let

$$S_n = \sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right)$$

Using part (a),

$$\begin{aligned} S_n &= \sum_{r=1}^n \left[ \ln \left( \frac{r+1}{r} \right) - \ln \left( \frac{r+3}{r+2} \right) \right] \\ &= \left( \ln 2 - \ln \frac{4}{3} \right) + \left( \ln \frac{3}{2} - \ln \frac{5}{4} \right) + \left( \ln \frac{4}{3} - \ln \frac{6}{5} \right) + \cdots + \left( \ln \frac{n+1}{n} - \ln \frac{n+3}{n+2} \right) \end{aligned}$$

Now write the positive and negative terms together:

$$\begin{aligned} S_n &= \ln 2 + \ln \frac{3}{2} + \cancel{\ln \frac{4}{3}} + \cancel{\ln \frac{3}{4}} + \cancel{\ln \frac{4}{5}} + \cdots + \cancel{\ln \frac{n+1}{n}} \\ &\quad - \cancel{\ln \frac{4}{3}} - \cancel{\ln \frac{5}{4}} - \cancel{\ln \frac{6}{5}} - \cdots - \cancel{\ln \frac{n+2}{n+1}} - \ln \frac{n+2}{n+1} - \ln \frac{n+3}{n+2} \end{aligned}$$

All the coloured middle terms cancel, so

$$\begin{aligned} S_n &= \ln 2 + \ln \frac{3}{2} - \ln \frac{n+2}{n+1} - \ln \frac{n+3}{n+2} \\ &= \ln \left( 2 \cdot \frac{3}{2} \cdot \frac{n+1}{n+2} \cdot \frac{n+2}{n+3} \right) \\ &= \ln \left( \frac{3(n+1)}{n+3} \right) \end{aligned}$$

Therefore,

$$\sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right) = \ln \left( \frac{3(n+1)}{n+3} \right)$$

2. Let  $a$  be a positive constant.

(a) Use the method of differences to find

$$\sum_{r=1}^n \frac{1}{(r+a)(r+a+1)(r+a+2)}$$

in terms of  $n$  and  $a$ .

[4]

(b) Hence find the value of  $a$  for which

$$\sum_{r=1}^{\infty} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{24}$$

[3]

### Solution

(a) Let

$$S_n = \sum_{r=1}^n \frac{1}{(r+a)(r+a+1)(r+a+2)}$$

To use the method of differences, first note that for any  $x$ ,

$$\begin{aligned} \frac{1}{x(x+1)} - \frac{1}{(x+1)(x+2)} &= \frac{x+2-x}{x(x+1)(x+2)} \\ &= \frac{2}{x(x+1)(x+2)} \end{aligned}$$

So

$$\frac{1}{x(x+1)(x+2)} = \frac{1}{2} \left( \frac{1}{x(x+1)} - \frac{1}{(x+1)(x+2)} \right)$$

Now put  $x = r + a$ . Then

$$S_n = \frac{1}{2} \sum_{r=1}^n \left( \frac{1}{(r+a)(r+a+1)} - \frac{1}{(r+a+1)(r+a+2)} \right)$$

Expanding the sum,

$$\begin{aligned} S_n &= \frac{1}{2} \left[ \frac{1}{(a+1)(a+2)} - \frac{1}{\cancel{(a+2)(a+3)}} \right. \\ &\quad + \frac{1}{\cancel{(a+2)(a+3)}} - \frac{1}{\cancel{(a+3)(a+4)}} \\ &\quad + \frac{1}{\cancel{(a+3)(a+4)}} - \dots \\ &\quad \left. + \frac{1}{\cancel{(n+a)(n+a+1)}} - \frac{1}{\cancel{(n+a)(n+a+1)}} - \frac{1}{(n+a+1)(n+a+2)} \right] \end{aligned}$$

All the middle terms cancel, leaving

$$S_n = \frac{1}{2} \left( \frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)} \right)$$

Hence

$$\sum_{r=1}^n \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{2} \left( \frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)} \right)$$

(b) From part (a),

$$\sum_{r=1}^n \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{2} \left( \frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)} \right)$$

As  $n \rightarrow \infty$ ,

$$\frac{1}{(n+a+1)(n+a+2)} \rightarrow 0$$

So

$$\sum_{r=1}^{\infty} \frac{1}{(r+a)(r+a+1)(r+a+2)} = \frac{1}{2(a+1)(a+2)}$$

Given that this equals  $\frac{1}{24}$ ,

$$\frac{1}{2(a+1)(a+2)} = \frac{1}{24}$$

$$2(a+1)(a+2) = 24$$

$$(a+1)(a+2) = 12$$

$$a^2 + 3a + 2 = 12$$

$$a^2 + 3a - 10 = 0$$

$$(a-2)(a+5) = 0$$

So  $a = 2$  or  $a = -5$ . Since  $a$  is positive,

$$a = 2$$

3. (a) Show that, for  $r \geq 2$ ,

$$\frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = \sqrt{r(r+2)} - \sqrt{r(r-2)} \quad [2]$$

(b) Hence use the method of differences to determine

$$\sum_{r=2}^n \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}}$$

giving your answer in simplest form. [3]

(c) Hence show that

$$\sum_{r=3}^7 \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = A\sqrt{2} + B\sqrt{3} + C\sqrt{7}$$

where  $A$ ,  $B$  and  $C$  are integers to be determined. [2]

### Solution

(a) Rationalise the denominator:

$$\begin{aligned} \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} &= \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} \cdot \frac{\sqrt{r(r+2)} - \sqrt{r(r-2)}}{\sqrt{r(r+2)} - \sqrt{r(r-2)}} \\ &= \frac{4r \left( \sqrt{r(r+2)} - \sqrt{r(r-2)} \right)}{r(r+2) - r(r-2)} \\ &= \frac{4r \left( \sqrt{r(r+2)} - \sqrt{r(r-2)} \right)}{4r} \\ &= \sqrt{r(r+2)} - \sqrt{r(r-2)} \end{aligned}$$

Hence

$$\frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = \sqrt{r(r+2)} - \sqrt{r(r-2)}$$

as required.

(b) Let

$$u_r = \sqrt{r(r+2)}$$

Then from part (a),

$$\frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = u_r - u_{r-2}$$

So, if

$$S_n = \sum_{r=2}^n \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}}$$

then

$$\begin{aligned} S_n &= \sum_{r=2}^n (u_r - u_{r-2}) \\ &= \sum_{r=2}^n u_r - \sum_{r=2}^n u_{r-2} \\ &= \sum_{r=2}^n u_r - \sum_{r=0}^{n-2} u_r \end{aligned}$$

using a change of index in the second sum.

Writing out the terms shows the method of differences clearly:

$$\begin{aligned} S_n &= (\cancel{u_2} + \cancel{u_3} + \cancel{u_4} + \cancel{u_5} + \cdots + \cancel{u_{n-2}} + u_{n-1} + u_n) \\ &\quad - (u_0 + u_1 + \cancel{u_2} + \cancel{u_3} + \cancel{u_4} + \cancel{u_5} + \cdots + \cancel{u_{n-2}}) \\ &= u_{n-1} + u_n - u_0 - u_1 \end{aligned}$$

Now

$$u_{n-1} = \sqrt{(n-1)(n+1)}, \quad u_n = \sqrt{n(n+2)}, \quad u_0 = \sqrt{0 \cdot 2} = 0, \quad u_1 = \sqrt{1 \cdot 3} = \sqrt{3}$$

Hence

$$\sum_{r=2}^n \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = \sqrt{(n-1)(n+1)} + \sqrt{n(n+2)} - \sqrt{3}$$

This is the required sum.

(c) Using part (b) with  $n = 7$ ,

$$\begin{aligned} \sum_{r=2}^7 \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} &= \sqrt{6 \cdot 8} + \sqrt{7 \cdot 9} - \sqrt{3} \\ &= \sqrt{48} + \sqrt{63} - \sqrt{3} \\ &= 4\sqrt{3} + 3\sqrt{7} - \sqrt{3} \\ &= 3\sqrt{3} + 3\sqrt{7} \end{aligned}$$

The  $r = 2$  term is

$$\frac{4(2)}{\sqrt{2(2+2)} + \sqrt{2(2-2)}} = \sqrt{2 \cdot 4} - \sqrt{2 \cdot 0} = \sqrt{8} = 2\sqrt{2}$$

from part (a).

Therefore

$$\begin{aligned} \sum_{r=3}^7 \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} &= \sum_{r=2}^7 \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} - 2\sqrt{2} \\ &= 3\sqrt{3} + 3\sqrt{7} - 2\sqrt{2} \end{aligned}$$

So

$$\sum_{r=3}^7 \frac{4r}{\sqrt{r(r+2)} + \sqrt{r(r-2)}} = -2\sqrt{2} + 3\sqrt{3} + 3\sqrt{7}$$

Hence

$$A = -2, \quad B = 3, \quad C = 3$$

4. Use the method of differences to show that

$$\sum_{r=1}^n \frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{n(an+b)}{(2n+1)(2n+3)}$$

where  $a$  and  $b$  are integers to be determined.

[6]

### Solution

Let

$$S_n = \sum_{r=1}^n \frac{12}{(2r-1)(2r+1)(2r+3)}$$

Write the general term in partial fractions:

$$\frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{A}{2r-1} + \frac{B}{2r+1} + \frac{C}{2r+3}$$

Using the cover-up rule:

$$A = \frac{12}{(2r+1)(2r+3)} \Big|_{2r-1=0} = \frac{12}{2 \cdot 4} = \frac{3}{2}$$

$$B = \frac{12}{(2r-1)(2r+3)} \Big|_{2r+1=0} = \frac{12}{(-2)(2)} = -3$$

$$C = \frac{12}{(2r-1)(2r+1)} \Big|_{2r+3=0} = \frac{12}{(-4)(-2)} = \frac{3}{2}$$

So

$$\frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{3}{2(2r-1)} - \frac{3}{2r+1} + \frac{3}{2(2r+3)}$$

It is convenient to rewrite this as

$$\frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{3}{2} \left( \frac{1}{2r-1} - \frac{1}{2r+1} \right) - \frac{3}{2} \left( \frac{1}{2r+1} - \frac{1}{2r+3} \right)$$

Hence

$$\begin{aligned} S_n &= \sum_{r=1}^n \frac{12}{(2r-1)(2r+1)(2r+3)} \\ &= \frac{3}{2} \sum_{r=1}^n \left( \frac{1}{2r-1} - \frac{1}{2r+1} \right) - \frac{3}{2} \sum_{r=1}^n \left( \frac{1}{2r+1} - \frac{1}{2r+3} \right) \end{aligned}$$

Expanding both telescoping sums:

$$\begin{aligned} S_n &= \frac{3}{2} \left[ \left( 1 - \frac{1}{3} \right) + \left( \frac{1}{3} - \frac{1}{5} \right) + \left( \frac{1}{5} - \frac{1}{7} \right) + \cdots + \left( \frac{1}{2n-1} - \frac{1}{2n+1} \right) \right] \\ &\quad - \frac{3}{2} \left[ \left( \frac{1}{3} - \frac{1}{5} \right) + \left( \frac{1}{5} - \frac{1}{7} \right) + \left( \frac{1}{7} - \frac{1}{9} \right) + \cdots + \left( \frac{1}{2n+1} - \frac{1}{2n+3} \right) \right] \end{aligned}$$

So

$$\begin{aligned} S_n &= \frac{3}{2} \left[ 1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \frac{1}{7} - \frac{1}{9} + \cdots - \frac{1}{2n+1} \right] \\ &\quad - \frac{3}{2} \left[ \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \frac{1}{7} - \frac{1}{9} + \cdots - \frac{1}{2n+3} \right] \end{aligned}$$

Hence

$$\begin{aligned} S_n &= \frac{3}{2} \left( 1 - \frac{1}{2n+1} \right) - \frac{3}{2} \left( \frac{1}{3} - \frac{1}{2n+3} \right) \\ &= \frac{3}{2} - \frac{3}{2(2n+1)} - \frac{1}{2} + \frac{3}{2(2n+3)} \\ &= 1 + \frac{3}{2} \left( \frac{1}{2n+3} - \frac{1}{2n+1} \right) \\ &= 1 - \frac{3}{(2n+1)(2n+3)} \\ &= \frac{(2n+1)(2n+3) - 3}{(2n+1)(2n+3)} \\ &= \frac{4n^2 + 8n}{(2n+1)(2n+3)} \\ &= \frac{n(4n+8)}{(2n+1)(2n+3)} \end{aligned}$$

Comparing with

$$\frac{n(an+b)}{(2n+1)(2n+3)}$$

gives

$$a = 4, \quad b = 8$$

So

$$\sum_{r=1}^n \frac{12}{(2r-1)(2r+1)(2r+3)} = \frac{n(4n+8)}{(2n+1)(2n+3)}$$

and the integers are  $a = 4$  and  $b = 8$ .

5. (a) Use the method of differences to prove that for  $n \geq 1$

$$\sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right) = \ln \left( \frac{3(n+1)}{n+3} \right) \quad [4]$$

- (b) Hence find the exact value of

$$\sum_{r=25}^{74} \ln \left( \left( 1 + \frac{2}{r(r+3)} \right)^{28} \right)$$

Give your answer in the form  $a \ln \left( \frac{b}{c} \right)$  where  $a$ ,  $b$  and  $c$  are integers to be determined. [3]

### Solution

- (a) Let

$$S_n = \sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right)$$

First simplify the term inside the logarithm:

$$\begin{aligned} 1 + \frac{2}{r(r+3)} &= \frac{r(r+3) + 2}{r(r+3)} \\ &= \frac{r^2 + 3r + 2}{r(r+3)} \\ &= \frac{(r+1)(r+2)}{r(r+3)} \\ &= \frac{r+1}{r} \cdot \frac{r+2}{r+3} \end{aligned}$$

So

$$\begin{aligned} \ln \left( 1 + \frac{2}{r(r+3)} \right) &= \ln \left( \frac{r+1}{r} \cdot \frac{r+2}{r+3} \right) \\ &= \ln(r+1) - \ln r + \ln(r+2) - \ln(r+3) \end{aligned}$$

Hence

$$S_n = \sum_{r=1}^n (\ln(r+1) - \ln r) + \sum_{r=1}^n (\ln(r+2) - \ln(r+3))$$

Now expand each sum using the method of differences.

For the first sum,

$$\begin{aligned} \sum_{r=1}^n (\ln(r+1) - \ln r) &= (\ln 2 + \ln 3 + \ln 4 + \cdots + \ln n + \ln(n+1)) \\ &\quad - (\ln 1 + \ln 2 + \ln 3 + \ln 4 + \cdots + \ln n) \\ &= \ln(n+1) - \ln 1 \end{aligned}$$

For the second sum,

$$\begin{aligned} \sum_{r=1}^n (\ln(r+2) - \ln(r+3)) &= (\ln 3 + \ln 4 + \ln 5 + \cdots + \ln(n+2)) \\ &\quad - (\ln 4 + \ln 5 + \cdots + \ln(n+2) + \ln(n+3)) \\ &= \ln 3 - \ln(n+3) \end{aligned}$$

Therefore

$$\begin{aligned} S_n &= (\ln(n+1) - \ln 1) + (\ln 3 - \ln(n+3)) \\ &= \ln(n+1) + \ln 3 - \ln(n+3) \end{aligned}$$

Since  $\ln 1 = 0$ ,

$$S_n = \ln \left( \frac{3(n+1)}{n+3} \right)$$

So

$$\sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right) = \ln \left( \frac{3(n+1)}{n+3} \right)$$

(b) Using  $\ln(a^{28}) = 28 \ln a$ ,

$$\sum_{r=25}^{74} \ln \left( \left( 1 + \frac{2}{r(r+3)} \right)^{28} \right) = 28 \sum_{r=25}^{74} \ln \left( 1 + \frac{2}{r(r+3)} \right)$$

From part (a), if

$$S_n = \sum_{r=1}^n \ln \left( 1 + \frac{2}{r(r+3)} \right)$$

then

$$S_n = \ln \left( \frac{3(n+1)}{n+3} \right)$$

So

$$\begin{aligned} \sum_{r=25}^{74} \ln \left( 1 + \frac{2}{r(r+3)} \right) &= S_{74} - S_{24} \\ &= \ln \left( \frac{3(75)}{77} \right) - \ln \left( \frac{3(25)}{27} \right) \\ &= \ln \left( \frac{225}{77} \right) - \ln \left( \frac{75}{27} \right) \\ &= \ln \left( \frac{225}{77} \cdot \frac{27}{75} \right) \\ &= \ln \left( \frac{81}{77} \right) \end{aligned}$$

Hence

$$\sum_{r=25}^{74} \ln \left( \left( 1 + \frac{2}{r(r+3)} \right)^{28} \right) = 28 \ln \left( \frac{81}{77} \right)$$

So the exact value is

$$28 \ln \left( \frac{81}{77} \right)$$

6. Using partial fractions and a method of differences, prove that for positive integer  $n$ ,

$$\sum_{r=1}^n \frac{6}{r(r+1)(r+2)} = \frac{n(an+b)}{2(n+1)(n+2)}$$

where  $a$  and  $b$  are constants to be found.

[6]

### Solution

Let

$$S_n = \sum_{r=1}^n \frac{6}{r(r+1)(r+2)}$$

Using partial fractions, write

$$\frac{6}{r(r+1)(r+2)} = \frac{A}{r(r+1)} + \frac{B}{(r+1)(r+2)}$$

Multiplying through by  $r(r+1)(r+2)$ ,

$$6 = A(r+2) + Br$$

So

$$6 = (A+B)r + 2A$$

Comparing coefficients gives

$$A+B=0, \quad 2A=6$$

Hence

$$A=3, \quad B=-3$$

Therefore

$$\frac{6}{r(r+1)(r+2)} = \frac{3}{r(r+1)} - \frac{3}{(r+1)(r+2)}$$

So

$$S_n = \sum_{r=1}^n \left( \frac{3}{r(r+1)} - \frac{3}{(r+1)(r+2)} \right)$$

Now use the method of differences by writing out the first few and last few terms:

$$\begin{aligned} S_n &= \left( \frac{3}{1 \cdot 2} - \frac{3}{2 \cdot 3} \right) + \left( \frac{3}{2 \cdot 3} - \frac{3}{3 \cdot 4} \right) + \left( \frac{3}{3 \cdot 4} - \frac{3}{4 \cdot 5} \right) + \dots \\ &\quad + \left( \frac{3}{(n-1)n} - \frac{3}{n(n+1)} \right) + \left( \frac{3}{n(n+1)} - \frac{3}{(n+1)(n+2)} \right) \end{aligned}$$

All the coloured pairs cancel, leaving

$$S_n = \frac{3}{1 \cdot 2} - \frac{3}{(n+1)(n+2)}$$

So

$$S_n = \frac{3}{2} - \frac{3}{(n+1)(n+2)}$$

Put this over the common denominator  $2(n+1)(n+2)$ :

$$\begin{aligned} S_n &= \frac{3(n+1)(n+2) - 6}{2(n+1)(n+2)} \\ &= \frac{3(n^2 + 3n + 2) - 6}{2(n+1)(n+2)} \\ &= \frac{3n^2 + 9n}{2(n+1)(n+2)} \\ &= \frac{3n(n+3)}{2(n+1)(n+2)} \end{aligned}$$

We are told this has the form

$$\frac{n(an+b)}{2(n+1)(n+2)}$$

Hence

$$n(an+b) = 3n(n+3) = n(3n+9)$$

Since  $n$  is a positive integer,  $n \neq 0$ , so

$$an+b = 3n+9$$

Therefore

$$a = 3, \quad b = 9$$

Hence

$$\sum_{r=1}^n \frac{6}{r(r+1)(r+2)} = \frac{3n(n+3)}{2(n+1)(n+2)} = \frac{n(3n+9)}{2(n+1)(n+2)}$$

so the required constants are  $a = 3$  and  $b = 9$ .

7. Let  $T_n = (n+1)^3 + (n+2)^3 + \dots + (2n)^3$ .

(a) By considering  $(r+1)^4 - (r-1)^4$ , use the method of differences to prove that

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2 \quad [5]$$

(b) Show that

$$T_n = \frac{1}{4}n^2(an^2 + bn + c)$$

where  $a$ ,  $b$  and  $c$  are integers to be determined. [3]

(c) State the value of

$$\lim_{n \rightarrow \infty} \frac{T_n}{n^4} \quad [1]$$

### Solution

(a) First expand

$$(r+1)^4 - (r-1)^4 = (r^4 + 4r^3 + 6r^2 + 4r + 1) - (r^4 - 4r^3 + 6r^2 - 4r + 1) = 8r^3 + 8r$$

Now sum from  $r = 1$  to  $n$ :

$$\sum_{r=1}^n ((r+1)^4 - (r-1)^4) = 8 \sum_{r=1}^n r^3 + 8 \sum_{r=1}^n r$$

Using the method of differences on the left-hand side,

$$\begin{aligned} \sum_{r=1}^n ((r+1)^4 - (r-1)^4) &= (2^4 + 3^4 + 4^4 + \dots + (n-1)^4 + n^4 + (n+1)^4) \\ &\quad - (0^4 + 1^4 + 2^4 + 3^4 + 4^4 + \dots + (n-1)^4) \\ &= (n+1)^4 + n^4 - 1 \end{aligned}$$

Therefore

$$(n+1)^4 + n^4 - 1 = 8 \sum_{r=1}^n r^3 + 8 \sum_{r=1}^n r$$

Using

$$\sum_{r=1}^n r = \frac{n(n+1)}{2}$$

we get

$$\begin{aligned} 8 \sum_{r=1}^n r^3 &= (n+1)^4 + n^4 - 1 - 8 \left( \frac{n(n+1)}{2} \right) \\ &= (n+1)^4 + n^4 - 1 - 4n(n+1) \\ &= (n^4 + 4n^3 + 6n^2 + 4n + 1) + n^4 - 1 - 4n^2 - 4n \\ &= 2n^4 + 4n^3 + 2n^2 \\ &= 2n^2(n+1)^2 \end{aligned}$$

So

$$\sum_{r=1}^n r^3 = \frac{2n^2(n+1)^2}{8} = \frac{1}{4}n^2(n+1)^2$$

Hence

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$$

(b)

$$T_n = (n+1)^3 + (n+2)^3 + \cdots + (2n)^3 = \sum_{r=n+1}^{2n} r^3$$

So

$$\begin{aligned} T_n &= \sum_{r=1}^{2n} r^3 - \sum_{r=1}^n r^3 \\ &= \frac{1}{4}(2n)^2(2n+1)^2 - \frac{1}{4}n^2(n+1)^2 \end{aligned}$$

Now simplify:

$$\begin{aligned} T_n &= \frac{1}{4} (4n^2(2n+1)^2 - n^2(n+1)^2) \\ &= \frac{1}{4} (4n^2(4n^2 + 4n + 1) - n^2(n^2 + 2n + 1)) \\ &= \frac{1}{4} (16n^4 + 16n^3 + 4n^2 - n^4 - 2n^3 - n^2) \\ &= \frac{1}{4} (15n^4 + 14n^3 + 3n^2) \\ &= \frac{1}{4} n^2 (15n^2 + 14n + 3) \end{aligned}$$

Therefore

$$T_n = \frac{1}{4}n^2(an^2 + bn + c) \quad \text{with} \quad a = 15, \quad b = 14, \quad c = 3$$

(c) From part (b),

$$\frac{T_n}{n^4} = \frac{1}{4} \left( 15 + \frac{14}{n} + \frac{3}{n^2} \right)$$

As  $n \rightarrow \infty$ ,  $\frac{14}{n} \rightarrow 0$  and  $\frac{3}{n^2} \rightarrow 0$ , so

$$\lim_{n \rightarrow \infty} \frac{T_n}{n^4} = \frac{15}{4}$$

8. (a) By expressing  $\frac{1}{r(r+2)}$  in partial fractions, show that

$$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)} \quad [5]$$

- (b) Hence determine the value of

$$\sum_{r=1}^{\infty} \frac{1}{r(r+2)} \quad [2]$$

### Solution

- (a) Let

$$S_n = \sum_{r=1}^n \frac{1}{r(r+2)}$$

First express the general term in partial fractions:

$$\frac{1}{r(r+2)} = \frac{A}{r} + \frac{B}{r+2}$$

So

$$1 = A(r+2) + Br = (A+B)r + 2A$$

Comparing coefficients gives

$$A+B=0, \quad 2A=1$$

Hence

$$A = \frac{1}{2}, \quad B = -\frac{1}{2}$$

Therefore

$$\frac{1}{r(r+2)} = \frac{1}{2} \left( \frac{1}{r} - \frac{1}{r+2} \right)$$

Substituting into the sum,

$$\begin{aligned} S_n &= \frac{1}{2} \sum_{r=1}^n \left( \frac{1}{r} - \frac{1}{r+2} \right) \\ &= \frac{1}{2} \left[ \left( 1 - \frac{1}{3} \right) + \left( \frac{1}{2} - \frac{1}{4} \right) + \left( \frac{1}{3} - \frac{1}{5} \right) + \cdots + \left( \frac{1}{n} - \frac{1}{n+2} \right) \right] \end{aligned}$$

Now use the method of differences:

$$\begin{aligned} S_n &= \frac{1}{2} \left( 1 + \frac{1}{2} + \frac{\cancel{1/3}}{\cancel{3}} + \frac{\cancel{1/4}}{\cancel{4}} + \cdots + \frac{\cancel{1/n}}{\cancel{n}} - \frac{\cancel{1/3}}{\cancel{3}} - \frac{\cancel{1/4}}{\cancel{4}} - \cdots - \frac{\cancel{1/n}}{\cancel{n}} - \frac{1}{n+1} - \frac{1}{n+2} \right) \\ &= \frac{1}{2} \left( 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right) \\ &= \frac{1}{2} \left( \frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right) \\ &= \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)} \end{aligned}$$

Hence

$$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}$$

(b) From part (a),

$$\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}$$

Letting  $n \rightarrow \infty$ ,

$$\frac{1}{2(n+1)} \rightarrow 0, \quad \frac{1}{2(n+2)} \rightarrow 0$$

so

$$\sum_{r=1}^{\infty} \frac{1}{r(r+2)} = \frac{3}{4}$$

The value of the infinite series is  $\frac{3}{4}$ .

9. Show that

$$\frac{1}{1 \times 4} + \frac{1}{2 \times 5} + \frac{1}{3 \times 6} + \cdots + \frac{1}{n(n+3)} = \frac{11}{18} - \frac{an^2 + bn + c}{3(n+1)(n+2)(n+3)}$$

where  $a$ ,  $b$  and  $c$  are constants to be found.

[6]

### Solution

Let

$$S = \frac{1}{1 \times 4} + \frac{1}{2 \times 5} + \frac{1}{3 \times 6} + \cdots + \frac{1}{n(n+3)} = \sum_{r=1}^n \frac{1}{r(r+3)}$$

Use partial fractions on the general term:

$$\frac{1}{r(r+3)} = \frac{A}{r} + \frac{B}{r+3}$$

So

$$1 = A(r+3) + Br = (A+B)r + 3A$$

Comparing coefficients:

$$A+B=0, \quad 3A=1$$

Hence

$$A = \frac{1}{3}, \quad B = -\frac{1}{3}$$

Therefore

$$\frac{1}{r(r+3)} = \frac{1}{3} \left( \frac{1}{r} - \frac{1}{r+3} \right)$$

So

$$S = \frac{1}{3} \sum_{r=1}^n \left( \frac{1}{r} - \frac{1}{r+3} \right)$$

Now write out the terms to use the method of differences:

$$S = \frac{1}{3} \left[ 1 + \frac{1}{2} + \frac{1}{3} + \frac{\cancel{\frac{1}{4}}}{4} + \frac{\cancel{\frac{1}{5}}}{5} + \cdots + \frac{\cancel{\frac{1}{n}}}{n} \right. \\ \left. - \frac{\cancel{\frac{1}{4}}}{4} - \frac{\cancel{\frac{1}{5}}}{5} - \cdots - \frac{\cancel{\frac{1}{n}}}{n} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} \right]$$

Everything from  $\frac{1}{4}$  to  $\frac{1}{n}$  cancels, leaving

$$S = \frac{1}{3} \left( 1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} \right)$$

Now

$$1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$$

Hence

$$S = \frac{11}{18} - \frac{1}{3} \left( \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} \right)$$

Combine the three fractions:

$$\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} = \frac{(n+2)(n+3) + (n+1)(n+3) + (n+1)(n+2)}{(n+1)(n+2)(n+3)}$$

Expand the numerator:

$$(n+2)(n+3) = n^2 + 5n + 6$$

$$(n+1)(n+3) = n^2 + 4n + 3$$

$$(n+1)(n+2) = n^2 + 3n + 2$$

So

$$(n+2)(n+3) + (n+1)(n+3) + (n+1)(n+2) = 3n^2 + 12n + 11$$

Therefore

$$S = \frac{11}{18} - \frac{3n^2 + 12n + 11}{3(n+1)(n+2)(n+3)}$$

Comparing with

$$\frac{11}{18} - \frac{an^2 + bn + c}{3(n+1)(n+2)(n+3)}$$

we get

$$a = 3, \quad b = 12, \quad c = 11$$

So the required result is

$$\frac{1}{1 \times 4} + \frac{1}{2 \times 5} + \frac{1}{3 \times 6} + \cdots + \frac{1}{n(n+3)} = \frac{11}{18} - \frac{3n^2 + 12n + 11}{3(n+1)(n+2)(n+3)}$$

10. (a) Use standard results from the formula booklet to show that

$$\sum_{r=1}^N r(2r-1)(2r+1) = \frac{1}{2}N(N+1)(2N^2+2N-1) \quad [3]$$

(b) Express  $\frac{6r+13}{(2r-1)(2r+3)}$  in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^N \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1}$$

in terms of  $N$ . [4]

(c) Deduce the value of

$$\sum_{r=1}^{\infty} \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1} \quad [1]$$

### Solution

(a) Using

$$(2r-1)(2r+1) = 4r^2 - 1$$

we get

$$r(2r-1)(2r+1) = r(4r^2 - 1) = 4r^3 - r$$

So

$$\sum_{r=1}^N r(2r-1)(2r+1) = 4 \sum_{r=1}^N r^3 - \sum_{r=1}^N r$$

From the formula book,

$$\sum_{r=1}^N r = \frac{N(N+1)}{2}, \quad \sum_{r=1}^N r^3 = \left(\frac{N(N+1)}{2}\right)^2$$

Hence

$$\begin{aligned} \sum_{r=1}^N r(2r-1)(2r+1) &= 4 \left(\frac{N(N+1)}{2}\right)^2 - \frac{N(N+1)}{2} \\ &= N^2(N+1)^2 - \frac{1}{2}N(N+1) \\ &= \frac{1}{2}N(N+1)(2N(N+1) - 1) \\ &= \frac{1}{2}N(N+1)(2N^2+2N-1) \end{aligned}$$

So

$$\sum_{r=1}^N r(2r-1)(2r+1) = \frac{1}{2}N(N+1)(2N^2+2N-1)$$

(b) Let

$$\frac{6r+13}{(2r-1)(2r+3)} = \frac{A}{2r-1} + \frac{B}{2r+3}$$

Then

$$6r+13 = A(2r+3) + B(2r-1)$$

so

$$6r + 13 = (2A + 2B)r + (3A - B)$$

Comparing coefficients,

$$2A + 2B = 6, \quad 3A - B = 13$$

so

$$A + B = 3, \quad 3A - B = 13$$

Adding gives  $4A = 16$ , so  $A = 4$ , and then  $B = -1$ .

Therefore

$$\frac{6r + 13}{(2r - 1)(2r + 3)} = \frac{4}{2r - 1} - \frac{1}{2r + 3}$$

Now let

$$S_N = \sum_{r=1}^N \frac{6r + 13}{(2r - 1)(2r + 3)} \left(\frac{1}{2}\right)^{r+1}$$

Then

$$\begin{aligned} S_N &= \sum_{r=1}^N \left( \frac{4}{2r - 1} - \frac{1}{2r + 3} \right) \frac{1}{2^{r+1}} \\ &= \sum_{r=1}^N \left( \frac{4}{(2r - 1)2^{r+1}} - \frac{1}{(2r + 3)2^{r+1}} \right) \\ &= \sum_{r=1}^N \left( \frac{4}{(2r - 1)2^{r+1}} - \frac{4}{(2r + 3)2^{r+3}} \right) \end{aligned}$$

Define

$$u_r = \frac{1}{(2r - 1)2^{r+1}}$$

Then

$$u_{r+2} = \frac{1}{(2r + 3)2^{r+3}}$$

so

$$S_N = 4 \sum_{r=1}^N (u_r - u_{r+2})$$

Using the method of differences,

$$\begin{aligned} S_N &= 4 \left[ (u_1 + u_2 + \textcolor{blue}{u_3} + \textcolor{red}{u_4} + \cdots + \textcolor{teal}{u_N}) \right. \\ &\quad \left. - (\textcolor{blue}{u_3} + \textcolor{red}{u_4} + \cdots + \textcolor{teal}{u_N} + u_{N+1} + u_{N+2}) \right] \\ &= 4(u_1 + u_2 - u_{N+1} - u_{N+2}) \end{aligned}$$

Now

$$u_1 = \frac{1}{1 \cdot 2^2} = \frac{1}{4}, \quad u_2 = \frac{1}{3 \cdot 2^3} = \frac{1}{24}$$

and

$$u_{N+1} = \frac{1}{(2N + 1)2^{N+2}}, \quad u_{N+2} = \frac{1}{(2N + 3)2^{N+3}}$$

So

$$\begin{aligned} S_N &= 4 \left( \frac{1}{4} + \frac{1}{24} - \frac{1}{(2N + 1)2^{N+2}} - \frac{1}{(2N + 3)2^{N+3}} \right) \\ &= \frac{7}{6} - \frac{1}{(2N + 1)2^N} - \frac{1}{(2N + 3)2^{N+1}} \end{aligned}$$

Combining the two final fractions,

$$\begin{aligned}\frac{1}{(2N+1)2^N} + \frac{1}{(2N+3)2^{N+1}} &= \frac{2}{(2N+1)2^{N+1}} + \frac{1}{(2N+3)2^{N+1}} \\ &= \frac{2(2N+3) + (2N+1)}{(2N+1)(2N+3)2^{N+1}} \\ &= \frac{6N+7}{(2N+1)(2N+3)2^{N+1}}\end{aligned}$$

Hence

$$\sum_{r=1}^N \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1} = \frac{7}{6} - \frac{1}{(2N+1)2^N} - \frac{1}{(2N+3)2^{N+1}}$$

or equivalently

$$\sum_{r=1}^N \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1} = \frac{7}{6} - \frac{6N+7}{(2N+1)(2N+3)2^{N+1}}$$

(c) From part (b),

$$S_N = \frac{7}{6} - \frac{6N+7}{(2N+1)(2N+3)2^{N+1}}$$

As  $N \rightarrow \infty$ , the second term tends to 0 because the denominator contains the factor  $2^{N+1}$ .

Therefore

$$\sum_{r=1}^{\infty} \frac{6r+13}{(2r-1)(2r+3)} \left(\frac{1}{2}\right)^{r+1} = \frac{7}{6}$$

11. (a) Show that

$$(2n)^5 - (2n-2)^5 \equiv 10(2n-1)^4 + 20(2n-1)^2 + 2 \quad [2]$$

(b) Hence, using the method of differences, show that for all positive integer values of  $n$ ,

$$\sum_{r=1}^n (2r-1)^4 = \frac{1}{15}n(2n-1)(2n+1)(an^2 + bn + c)$$

where  $a$ ,  $b$  and  $c$  are integers to be determined. [7]

### Solution

(a) Let  $m = 2n - 1$ . Then

$$2n = m + 1, \quad 2n - 2 = m - 1$$

so

$$\begin{aligned} (2n)^5 - (2n-2)^5 &= (m+1)^5 - (m-1)^5 \\ &= (m^5 + 5m^4 + 10m^3 + 10m^2 + 5m + 1) \\ &\quad - (m^5 - 5m^4 + 10m^3 - 10m^2 + 5m - 1) \\ &= 10m^4 + 20m^2 + 2 \end{aligned}$$

Substituting back  $m = 2n - 1$ ,

$$(2n)^5 - (2n-2)^5 = 10(2n-1)^4 + 20(2n-1)^2 + 2$$

So the identity is shown.

(b) Let

$$S = \sum_{r=1}^n (2r-1)^4$$

Using the result from part (a) with  $n$  replaced by  $r$ ,

$$(2r)^5 - (2r-2)^5 = 10(2r-1)^4 + 20(2r-1)^2 + 2$$

Now sum from  $r = 1$  to  $n$ :

$$\sum_{r=1}^n ((2r)^5 - (2r-2)^5) = 10 \sum_{r=1}^n (2r-1)^4 + 20 \sum_{r=1}^n (2r-1)^2 + 2n$$

That is,

$$\sum_{r=1}^n ((2r)^5 - (2r-2)^5) = 10S + 20 \sum_{r=1}^n (2r-1)^2 + 2n$$

For the left-hand side, use the method of differences:

$$\begin{aligned} \sum_{r=1}^n ((2r)^5 - (2r-2)^5) &= \left( \textcolor{blue}{2}^5 + \textcolor{red}{4}^5 + \textcolor{teal}{6}^5 + \cdots + \textcolor{violet}{(2n-2)}^5 + (2n)^5 \right) \\ &\quad - \left( 0^5 + \textcolor{blue}{2}^5 + \textcolor{red}{4}^5 + \textcolor{teal}{6}^5 + \cdots + \textcolor{violet}{(2n-2)}^5 \right) \\ &= (2n)^5 \\ &= 32n^5 \end{aligned}$$

So

$$32n^5 = 10S + 20 \sum_{r=1}^n (2r-1)^2 + 2n$$

Now find  $\sum_{r=1}^n (2r-1)^2$ :

$$\begin{aligned}
 \sum_{r=1}^n (2r-1)^2 &= \sum_{r=1}^n (4r^2 - 4r + 1) \\
 &= 4 \sum_{r=1}^n r^2 - 4 \sum_{r=1}^n r + \sum_{r=1}^n 1 \\
 &= 4 \cdot \frac{n(n+1)(2n+1)}{6} - 4 \cdot \frac{n(n+1)}{2} + n \\
 &= \frac{2n(n+1)(2n+1)}{3} - 2n(n+1) + n \\
 &= \frac{2n(n+1)(2n+1) - 6n(n+1) + 3n}{3} \\
 &= \frac{n(2(n+1)(2n+1) - 6(n+1) + 3)}{3} \\
 &= \frac{n(4n^2 - 1)}{3} \\
 &= \frac{n(2n-1)(2n+1)}{3}
 \end{aligned}$$

Substitute this into the equation for  $S$ :

$$32n^5 = 10S + 20 \left( \frac{n(2n-1)(2n+1)}{3} \right) + 2n$$

Hence

$$\begin{aligned}
 10S &= 32n^5 - \frac{20n(2n-1)(2n+1)}{3} - 2n \\
 &= 32n^5 - \frac{20n(4n^2 - 1)}{3} - 2n \\
 &= 32n^5 - \frac{80}{3}n^3 + \frac{20}{3}n - 2n \\
 &= 32n^5 - \frac{80}{3}n^3 + \frac{14}{3}n
 \end{aligned}$$

So

$$\begin{aligned}
 S &= \frac{1}{10} \left( 32n^5 - \frac{80}{3}n^3 + \frac{14}{3}n \right) \\
 &= \frac{48n^5 - 40n^3 + 7n}{15}
 \end{aligned}$$

Now factorise:

$$\begin{aligned}
 48n^5 - 40n^3 + 7n &= n(48n^4 - 40n^2 + 7) \\
 &= n(4n^2 - 1)(12n^2 - 7) \\
 &= n(2n-1)(2n+1)(12n^2 - 7)
 \end{aligned}$$

Therefore

$$\sum_{r=1}^n (2r-1)^4 = \frac{1}{15} n(2n-1)(2n+1)(12n^2 - 7)$$

Comparing with

$$\sum_{r=1}^n (2r-1)^4 = \frac{1}{15} n(2n-1)(2n+1)(an^2 + bn + c)$$

gives

$$a = 12, \quad b = 0, \quad c = -7$$

12. Let

$$S_n = \sum_{r=1}^n \frac{1}{(3r-2)(3r+4)}$$

where  $n \geq 1$ .

(a) Use the method of differences to show that

$$S_n = \frac{n(15n+17)}{8(3n+1)(3n+4)} \quad [6]$$

(b) Show that, for any number  $k$  greater than  $\frac{24}{5}$ , if the difference between  $\frac{5}{24}$  and  $S_n$  is less than  $\frac{1}{k}$ , then

$$n > \frac{k-15+\sqrt{k^2+81}}{18} \quad [6]$$

### Solution

(a) Write

$$\frac{1}{(3r-2)(3r+4)} = \frac{A}{3r-2} + \frac{B}{3r+4}$$

Then

$$1 = A(3r+4) + B(3r-2)$$

Comparing coefficients of  $r$  and constants gives

$$3A + 3B = 0, \quad 4A - 2B = 1$$

So

$$A + B = 0 \implies B = -A$$

and hence

$$4A - 2(-A) = 1 \implies 6A = 1 \implies A = \frac{1}{6}, \quad B = -\frac{1}{6}$$

Therefore

$$\frac{1}{(3r-2)(3r+4)} = \frac{1}{6} \left( \frac{1}{3r-2} - \frac{1}{3r+4} \right)$$

So

$$S_n = \frac{1}{6} \sum_{r=1}^n \left( \frac{1}{3r-2} - \frac{1}{3r+4} \right)$$

Now expand the sum:

$$S_n = \frac{1}{6} \left[ \left( 1 - \frac{1}{7} \right) + \left( \frac{1}{4} - \frac{1}{10} \right) + \left( \frac{1}{7} - \frac{1}{13} \right) + \left( \frac{1}{10} - \frac{1}{16} \right) + \dots \right. \\ \left. + \left( \frac{1}{3n-5} - \frac{1}{3n+1} \right) + \left( \frac{1}{3n-2} - \frac{1}{3n+4} \right) \right]$$

Rearranging to show the cancellation pattern,

$$S_n = \frac{1}{6} \left( 1 + \frac{1}{4} + \cancel{\frac{1}{7}} + \cancel{\frac{1}{10}} + \dots + \cancel{\frac{1}{3n-2}} \right. \\ \left. - \cancel{\frac{1}{7}} - \cancel{\frac{1}{10}} - \dots - \cancel{\frac{1}{3n-2}} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$$

The coloured middle terms cancel, leaving

$$S_n = \frac{1}{6} \left( 1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$$

Now simplify:

$$\begin{aligned} S_n &= \frac{1}{6} \left( \frac{5}{4} - \frac{(3n+4) + (3n+1)}{(3n+1)(3n+4)} \right) \\ S_n &= \frac{1}{6} \left( \frac{5}{4} - \frac{6n+5}{(3n+1)(3n+4)} \right) \end{aligned}$$

Putting the terms inside the bracket over a common denominator,

$$\begin{aligned} S_n &= \frac{1}{6} \left( \frac{5(3n+1)(3n+4) - 4(6n+5)}{4(3n+1)(3n+4)} \right) \\ &= \frac{1}{6} \left( \frac{5(9n^2 + 15n + 4) - 24n - 20}{4(3n+1)(3n+4)} \right) \\ &= \frac{1}{6} \left( \frac{45n^2 + 75n + 20 - 24n - 20}{4(3n+1)(3n+4)} \right) \\ &= \frac{1}{6} \left( \frac{45n^2 + 51n}{4(3n+1)(3n+4)} \right) \\ &= \frac{1}{6} \left( \frac{3n(15n + 17)}{4(3n+1)(3n+4)} \right) \end{aligned}$$

Hence

$$S_n = \frac{n(15n+17)}{8(3n+1)(3n+4)}$$

(b) From part (a),

$$S_n = \frac{1}{6} \left( \frac{5}{4} - \frac{6n+5}{(3n+1)(3n+4)} \right) = \frac{5}{24} - \frac{6n+5}{6(3n+1)(3n+4)}$$

So

$$\frac{5}{24} - S_n = \frac{6n+5}{6(3n+1)(3n+4)}$$

If this difference is less than  $\frac{1}{k}$ , then

$$\frac{6n+5}{6(3n+1)(3n+4)} < \frac{1}{k}$$

Since  $k > \frac{24}{5} > 0$ , all quantities are positive, so we may cross-multiply:

$$k(6n+5) < 6(3n+1)(3n+4)$$

Expanding:

$$6kn + 5k < 6(9n^2 + 15n + 4) = 54n^2 + 90n + 24$$

Hence

$$54n^2 + (90 - 6k)n + (24 - 5k) > 0$$

Now solve the corresponding quadratic equation

$$54n^2 + (90 - 6k)n + (24 - 5k) = 0$$

Using the quadratic formula,

$$n = \frac{-(90 - 6k) \pm \sqrt{(90 - 6k)^2 - 4(54)(24 - 5k)}}{108}$$

Simplify the discriminant:

$$\begin{aligned}(90 - 6k)^2 - 4(54)(24 - 5k) &= 36(15 - k)^2 - 216(24 - 5k) \\ &= 36((15 - k)^2 - 6(24 - 5k)) \\ &= 36(k^2 - 30k + 225 - 144 + 30k) \\ &= 36(k^2 + 81)\end{aligned}$$

Therefore

$$\begin{aligned}n &= \frac{6k - 90 \pm 6\sqrt{k^2 + 81}}{108} \\ &= \frac{k - 15 \pm \sqrt{k^2 + 81}}{18}\end{aligned}$$

Since the coefficient of  $n^2$  is positive, the inequality

$$54n^2 + (90 - 6k)n + (24 - 5k) > 0$$

holds for values of  $n$  outside the two roots.

Also,

$$k - 15 - \sqrt{k^2 + 81} < k - 15 - k = -15 < 0$$

So the smaller root is negative. As  $n \geq 1$ , it follows that

$$n > \frac{k - 15 + \sqrt{k^2 + 81}}{18}$$

This is the required result.